ASSIGNAMENT OS_LAB_03

Course code: CC-311 LAB

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Linux Commands

ls ~:

Lists all files and directories in your home directory.

ls –a:

Lists all files and directories, including hidden ones.

ls –f:

Lists files without sorting; shows directories with a / at the end and disables colorization.

ls –i:

Displays the inode number of each file along with its name.

ls –l:

Lists files in long format showing permissions, owner, group, size, and modification date.

```
ramsha@myVM:~$ ls ~
Desktop    hello.c       myfile.txt  ramsha          testdir
Documents  hello.c.save  mywork      randomfile.txt  textdir
Downloads  Music         Pictures    snap            txtdir
hello      myfile2.txt   Public      Templates       Videos
```

```
ramsha@myVM:~$ ls -a
.              Desktop       .local       .profile                  Templates
..             Documents     .mozilla     Public                    testdir
.bash_history  Downloads     Music        ramsha                    textdir
.bash_logout   .gnupg        myfile2.txt  randomfile.txt            txtdir
.bashrc        hello         myfile.txt   snap                      Videos
.cache         hello.c       mywork       .ssh
.config        hello.c.save  Pictures     .sudo_as_admin_successful
```

```
ramsha@myVM:~$ ls -f
myfile.txt  txtdir        .profile                  Videos
ramsha      .bashrc       myfile2.txt               .config
testdir     .ssh          hello                     .bash_history
.mozilla    .bash_logout  .sudo_as_admin_successful textdir
Downloads   Pictures      .                         Templates
Desktop     hello.c       .gnupg                    mywork
Public      ..            Documents
Music       .local        randomfile.txt
snap        hello.c.save  .cache
```

```
ramsha@myVM:~$ ls -i
408437 Desktop    410358 hello.c.save  408443 Pictures       408439 Templates
408441 Documents  408442 Music         408440 Public         410362 testdir
408438 Downloads  410343 myfile2.txt   552262 ramsha         410337 textdir
393509 hello      410365 myfile.txt    410364 randomfile.txt 410363 txtdir
411037 hello.c    410359 mywork        410330 snap           408444 Videos
```

```
ramsha@myVM:~$ ls -l
total 92
drwxr-xr-x 2 ramsha ramsha  4096 Oct  9 21:47 Desktop
drwxr-xr-x 2 ramsha ramsha  4096 Oct  9 21:47 Documents
drwxr-xr-x 2 ramsha ramsha  4096 Oct  9 21:47 Downloads
-rwxrwxr-x 1 ramsha ramsha 16696 Oct 24 07:49 hello
-rw-rw-r-- 1 ramsha ramsha    58 Oct 24 07:42 hello.c
-rw-rw-r-- 1 ramsha ramsha    57 Oct 24 06:38 hello.c.save
drwxr-xr-x 2 ramsha ramsha  4096 Oct  9 21:47 Music
-rw-rw-r-- 1 ramsha ramsha    24 Oct 10 22:02 myfile2.txt
-rw-rw-r-- 1 ramsha ramsha    48 Oct 17 23:25 myfile.txt
drwxrwxr-x 2 ramsha ramsha  4096 Oct 18 00:30 mywork
drwxr-xr-x 2 ramsha ramsha  4096 Oct  9 21:47 Pictures
drwxr-xr-x 2 ramsha ramsha  4096 Oct  9 21:47 Public
drwxrwxr-x 2 ramsha ramsha  4096 Oct 23 21:05 ramsha
-rw-rw-r-- 1 ramsha ramsha     0 Oct 10 22:14 randomfile.txt
drwx------ 3 ramsha ramsha  4096 Oct 23 21:04 snap
drwxr-xr-x 2 ramsha ramsha  4096 Oct  9 21:47 Templates
drwxrwxr-x 2 ramsha ramsha  4096 Oct 10 22:08 testdir
-rw-rw-r-- 1 ramsha ramsha    23 Oct 10 22:11 textdir
-rw-rw-r-- 1 ramsha ramsha    23 Oct 10 22:10 txtdir
drwxr-xr-x 2 ramsha ramsha  4096 Oct  9 21:47 Videos
```

Mkdir:

Creates a directory named laptop in the current location.

Mkdir –p keybord:

Creates keyboard and its subdirectory mouse; -p ensures parent directories are created if they don't exist.

Mkdir –m:

Creates a directory keyboard with permissions set to 755.

Mkdir –p –m:

Creates keyboard and mouse (parent directories if needed) with permissions 755.

mkdir -m u=rwx,g=rx,o=rx keyboard:

Creates directory keyboard with permissions: owner = rwx, group = rx, others = rx.

ls:

Lists files and directories in the current directory.

ls –ld keyboard:

Shows detailed information (permissions, owner, size, date) of keyboard directory itself, not its contents.

```
ramsha@myVM:~$ mkdir laptop
ramsha@myVM:~$ mkdirv-p keyboard
mkdirv-p: command not found
ramsha@myVM:~$ mkdir -m 755 keyboard
ramsha@myVM:~$ mkdir -p -m 755 keyboard
ramsha@myVM:~$ ls
Desktop    hello.c       Music        Pictures        snap       txtdir
Documents  hello.c.save  myfile2.txt  Public          Templates  Videos
Downloads  keyboard      myfile.txt   ramsha          testdir
hello      laptop        mywork       randomfile.txt  textdir
ramsha@myVM:~$ ls -ld keyboard
drwxr-xr-x 2 ramsha ramsha 4096 Oct 25 07:23 keyboard
```

chmod:

Changes permissions of keyboard to 754.

Umask:

Displays the current default permission mask for newly created files/directories.

umask:

Sets the default permission mask so new files/directories are not writable by group/others.

```
ramsha@myVM:~$ chmod 754 keyboard
ramsha@myVM:~$ umask
0002
ramsha@myVM:~$ umask 0222
ramsha@myVM:~$ ls -ld keyboard
drwxr-xr-- 2 ramsha ramsha 4096 Oct 25 07:23 keyboard
```

wc:

Shows word count, line count, and byte/character count of file.

diff:

Compares two files line by line and shows differences.

cat:

Displays the content of file1.txt on the terminal.

Nano:

Opens file.txt in Nano editor for editing.

diff:

Compares two files line by line and shows differences.

uniq:

Removes consecutive duplicate lines from file.

```
ramsha@myVM:~$ wc myfile1.txt
wc: myfile1.txt: No such file or directory
ramsha@myVM:~$ diff myfile2.txt file.txt
diff: file.txt: No such file or directory
ramsha@myVM:~$ cat myfile2.txt
This is my second file.
ramsha@myVM:~$ nano myfile2.txt
ramsha@myVM:~$ diff myfile2.txt randomfile.txt 1c1
diff: extra operand '1c1'
diff: Try 'diff --help' for more information.
ramsha@myVM:~$ diff myfile2.txt randomfile.txt
1d0
< This is my second file.
ramsha@myVM:~$ unique myfile2.txt

Command 'unique' not found, but can be installed with:

sudo apt install john

ramsha@myVM:~$ uniq myfile2.txt
This is my second file.
ramsha@myVM:~$ touch myfile1.txt
ramsha@myVM:~$ echo "this is my first file"
this is my first file
```

Cut:

The **c**ut command is used to extract specific sections of text from each line of a file or command output.
It's mainly used to get selected columns, characters, or fields from structured data.

Paste:

The paste command is used to merge lines from two or more files horizontally (side by side).
It joins corresponding lines from each file and separates them with a tab (by default).

```
ramsha@myVM:~$ sort myfile2.txt
This is my second file.
ramsha@myVM:~$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
irc:39
gnats:41
nobody:65534
systemd-network:100
systemd-resolve:101
systemd-timesync:102
messagebus:103
syslog:104
 apt:105
```

```
tss:106
uuidd:107
tcpdump:108
avahi-autoipd:109
usbmux:110
rtkit:111
dnsmasq:112
cups-pk-helper:113
speech-dispatcher:114
avahi:115
kernoops:116
saned:117
nm-openvpn:118
hplip:119
whoopsie:120
colord:121
geoclue:122
pulse:123
gnome-initial-setup:124
gdm:125
ramsha:1000
systemd-coredump:999
ramsha@myVM:~$ paste myfile2.txt randomfile.txt
This is my second file.
ramsha@myVM:~$ cat randomfile.txt
ramsha@myVM:~$ nano randomfile.txt
ramsha@myVM:~$ cat randomfile.txt
hii you are using linux
```

Compress:

The compress command in Linux is used to reduce the size of files by compressing their contents.

After compression, a new file is created with the **". Z"** extension.

```
ramsha@myVM:~$ compress largefile.txt

Command 'compress' not found, but can be installed with:

sudo apt install ncompress

ramsha@myVM:~$
ramsha@myVM:~$ which compress
ramsha@myVM:~$ sudo apt update
[sudo] password for ramsha:
Hit:1 http://us.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metada
ta [74.8 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 DEP-11
Metadata [208 B]
Get:6 http://security.ubuntu.com/ubuntu focal-security/universe amd64 DEP-11 Me
tadata [160 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 DEP-11
Metadata [940 B]
Get:8 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:9 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 DEP-11 Metad
ata [276 kB]
Get:10 http://us.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 DEP-1
1 Metadata [212 B]
```

```
Get:16 http://us.archive.ubuntu.com/ubuntu focal-backports/multiverse amd64 DEP
-11 Metadata [212 B]
Fetched 1,344 kB in 1min 10s (19.2 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
590 packages can be upgraded. Run 'apt list --upgradable' to see them.
ramsha@myVM:~$ sudo apt install ncompress -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  ncompress
0 upgraded, 1 newly installed, 0 to remove and 590 not upgraded.
Need to get 21.4 kB of archives.
After this operation, 63.5 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu focal/universe amd64 ncompress amd64
4.2.4.6-1 [21.4 kB]
Fetched 21.4 kB in 3s (7,923 B/s)
Selecting previously unselected package ncompress.
(Reading database ... 183280 files and directories currently installed.)
Preparing to unpack .../ncompress_4.2.4.6-1_amd64.deb ...
Unpacking ncompress (4.2.4.6-1) ...
Setting up ncompress (4.2.4.6-1) ...
Processing triggers for man-db (2.9.1-1) ...
ramsha@myVM:~$ sudo dnf install ncompress -y
sudo: dnf: command not found
```

```
ramsha@myVM:~$ which compress
/usr/bin/compress
ramsha@myVM:~$ compress myfile2.txt
ramsha@myVM:~$ ls
 Desktop        keyboard                    mywork          testdir
 Documents      laptop                      Pictures        textdir
 Downloads      Music                       Public          txtdir
 file.txt       myfile1.txt                 ramsha          Videos
 hello          myfile2.txt                 randomfile.txt
 hello.c       'myfile2.txt newfile.txt'    snap
 hello.c.save   myfile.txt                  Templates
ramsha@myVM:~$ bash
ramsha@myVM:~$ compress myfile2.txt
ramsha@myVM:~$ ls
 Desktop        keyboard                    mywork          testdir
 Documents      laptop                      Pictures        textdir
 Downloads      Music                       Public          txtdir
 file.txt       myfile1.txt                 ramsha          Videos
 hello          myfile2.txt                 randomfile.txt
 hello.c       'myfile2.txt newfile.txt'    snap
 hello.c.save   myfile.txt                  Templates
ramsha@myVM:~$ ls -l myfile2.txt
-rw-rw-r-- 1 ramsha ramsha 24 Oct 10 22:02 myfile2.txt
ramsha@myVM:~$ chmod u+w myfile2.txt
ramsha@myVM:~$ compress -f myfile2.txt
ramsha@myVM:~$ ls myfile2*
'myfile2.txt newfile.txt'   myfile2.txt.Z
```

Uncompress:

The Uncompre**ss** command in Linux is used to restore files that were compressed using the compress.

```
ramsha@myVM:~$ compress myfile.txt
ramsha@myVM:~$ ls
 Desktop        keyboard                    mywork          testdir
 Documents      laptop                      Pictures        textdir
 Downloads      Music                       Public          txtdir
 file.txt       myfile1.txt                 ramsha          Videos
 hello         'myfile2.txt newfile.txt'    randomfile.txt
 hello.c        myfile2.txt.Z               snap
 hello.c.save   myfile.txt                  Templates
ramsha@myVM:~$ uncompress myfile2.txt
ramsha@myVM:~$ ls
 Desktop        keyboard                    mywork          testdir
 Documents      laptop                      Pictures        textdir
 Downloads      Music                       Public          txtdir
 file.txt       myfile1.txt                 ramsha          Videos
 hello          myfile2.txt                 randomfile.txt
 hello.c       'myfile2.txt newfile.txt'    snap
 hello.c.save   myfile.txt                  Templates
```

gzip:

The gzip command in Linux is used to compress files to reduce their size and save disk space. It is faster and provides better compression compared to the traditional compress command. It having “.gz” extension after zipping.

gunzip:

The gunzip command in Linux is used to decompress files that were compressed using the gzip.

```
ramsha@myVM:~$ gzip myfile2.txt
ramsha@myVM:~$ ls
 Desktop        keyboard                    mywork          testdir
 Documents      laptop                      Pictures        textdir
 Downloads      Music                       Public          txtdir
 file.txt       myfile1.txt                 ramsha          Videos
 hello          myfile2.txt.gz              randomfile.txt
 hello.c       'myfile2.txt newfile.txt'    snap
 hello.c.save   myfile.txt                  Templates
ramsha@myVM:~$ gunzip myfile2.txt
ramsha@myVM:~$ ls
 Desktop        keyboard                    mywork          testdir
 Documents      laptop                      Pictures        textdir
 Downloads      Music                       Public          txtdir
 file.txt       myfile1.txt                 ramsha          Videos
 hello          myfile2.txt                 randomfile.txt
 hello.c       'myfile2.txt newfile.txt'    snap
 hello.c.save   myfile.txt                  Templates
```

grep:

The grep command in Linux is used to search for specific patterns or words within files or command outputs.

egrep, fgrep:

The egrep and fgrep commands are variations of the grep command used for pattern searching in files or text.

```
ramsha@myVM:~$ grep "hii" myfile1.txt myfile2.txt randomfile.txt
grep: myfile2.txt: No such file or directory
randomfile.txt:hii you are using linux
ramsha@myVM:~$ egrap "you|are" randomfile.txt

Command 'egrap' not found, did you mean:

  command 'egrep' from deb grep (3.4-1)
  command 'grap' from deb grap (1.45-1build1)

Try: sudo apt install <deb name>

ramsha@myVM:~$ egrep "you|are" randomfile.txt
hii you are using linux
ramsha@myVM:~$ cat myfile2.txt
cat: myfile2.txt: No such file or directory
ramsha@myVM:~$ pwd
/home/ramsha
ramsha@myVM:~$ find/ -name "myfile2.txt" 2>/div/null
bash: /div/null: No such file or directory
```